

Lab 2 Report

Proportional Control

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1 Introduction

1.1 Overview

The Kitticopter laboratory combines a physical analogue controller with a simulated vertical-motion plant through a data-acquisition (DAQ) interface. Session 1 identifies the plant from an applied voltage step. Session 2 uses that model to design, implement and test an altitude controller on Veroboard. The modelling, identification, design and evaluation below follow this experimental sequence [2].

1.2 Aim

The aim is to regulate altitude using proportional control while satisfying the four requirements in the 2026 laboratory brief [1]. Table 1.1 translates these requirements into quantities that can be checked against the measured and simulated responses.

Table 1.1: Laboratory requirements and their evaluation measures.

Requirement	Evaluation measure
Tracking accuracy $> 92\%$	$100(1 - \lvert x - y_\infty\rvert/\lvert x\rvert) > 92\%$ for a nonzero step reference.
Overshoot $< 30\%$	$M_p = 100(y_{\max} - y_\infty)/(y_\infty - y_0) < 30\%$.
Aerodynamic uncertainty $\pm 10\%$	Vary the identified drag coefficient b by $\pm 10\%$, keeping the thrust slope fixed.
Component tolerance $\pm 10\%$	Examine changes in loop gain, reference/feedback ratio and hover bias.

1.3 Objectives

1. Derive the velocity and position models from the free-body diagram and identify the unknown drag, thrust and sensor parameters.
2. Differentiate the recorded altitude, calculate the velocity gain and time constant, calibrate the sensor and validate the resulting model.
3. Select and justify the proportional gain using root-locus and time-response calculations; check the nominal 5 m step and uncertainty cases.
4. Document the three-stage controller circuit and assess the experimental response against the requirements.

The analysis uses the supplied identification CSV, circuit design files, Simulink images, board photograph and controller-test display. Photographs and original screenshots are collected in the appendices. All numerical plots in the main report are regenerated from these records or from the stated equations. One identification run and one experimental controller response are retained; repeated-trial averages are therefore not claimed. Reproducible code and worked calculations are available in the GitHub repository.

2 System Modelling

2.1 Methodology

The laboratory permits a unit-mass model with velocity-proportional aerodynamic drag and constant gravity [1]. Upward altitude is positive. Newton's second law is applied first, the actuator is linearised around hover, and the Laplace transform then gives a first-order velocity model. Integrating velocity produces the position model used for control.

2.2 Modelling

2.2.1 Free-Body and Block Diagrams

The forces in Figure 2.1 are upward rotor thrust, downward weight and drag opposing motion. The complete signal-flow diagram is given in Figure 2.2; the notation is defined in Table 2.1.

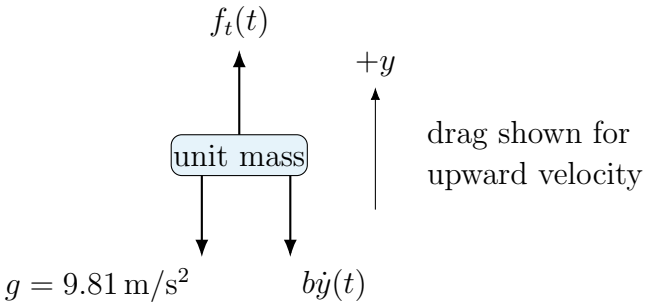


Figure 2.1: Free-body diagram of the normalised Kitticopter. Drag reverses direction when velocity reverses.

Table 2.1: Model notation and units.

Symbol	Description	Unit
$x, y, v = \dot{y}$	Reference altitude, altitude, vertical velocity	m, m, m/s
f_t, g, d_a	Mass-normalised thrust, gravity, extra disturbance	$\mathrm{m/s^2}$
b	Mass-normalised aerodynamic coefficient	$\mathrm{s^{-1}}$
$u, u_h, \delta u$	Command, hover command, $u - u_h$	V
k_T	Local thrust slope with respect to voltage	$\mathrm{m/(s^2\ V)}$
A, τ	Velocity gain and time constant	m/(s V), s
k_s	Position-to-voltage sensor coefficient	V/m
r_v, y_v, e_v	Reference, sensor, error voltage	V
K_p	Controller gain applied to voltage error	V/V

The unknowns required from identification are A, τ and k_s, from which b and k_T follow. The hover command is $u_h = 2.5$ V, the midpoint of the specified 0–5 V plant-input range.

2.2.2 Dynamic Model

Newton's second law, with mass normalised to one, gives

$$\ddot{y} = f_t - b\dot{y} - g + d_a. \tag{2.1}$$

At hover $f_{t,h} = g$. The local actuator relationship is $f_t = g + k_T(u - u_h)$; substituting this into Equation (2.1) gives

$$\ddot{y} + b\dot{y} = k_T\delta u + d_a, \qquad \delta u = u - u_h. \tag{2.2}$$

The hover operating point does not imply $k_T = g/u_h$: the slope must be obtained from the step response. With zero initial conditions and disturbances set to zero,

$$G_{vf}(s) = \frac{V(s)}{F_t(s)} = \frac{1}{s+b}, \tag{2.3}$$

$$G_v(s) = \frac{V(s)}{\Delta U(s)} = \frac{k_T}{s+b} = \frac{A}{\tau s+1}, \quad \tau = \frac{1}{b}, \quad A = \frac{k_T}{b}, \tag{2.4}$$

$$G_y(s) = \frac{Y(s)}{\Delta U(s)} = \frac{G_v(s)}{s} = \frac{A}{s(\tau s+1)}. \tag{2.5}$$

Thus the velocity response is first order and the position plant is second order. The integrator explains why a nonzero voltage deviation produces an altitude ramp after the velocity transient.

The reference is converted to voltage as $r_v = k_s x$, while the sensor gives $y_v = k_s y$. The implemented control law is

$$u = \mathrm{sat}_{[0,5]}\{2.5 + K_p(r_v - y_v)\}. \tag{2.6}$$

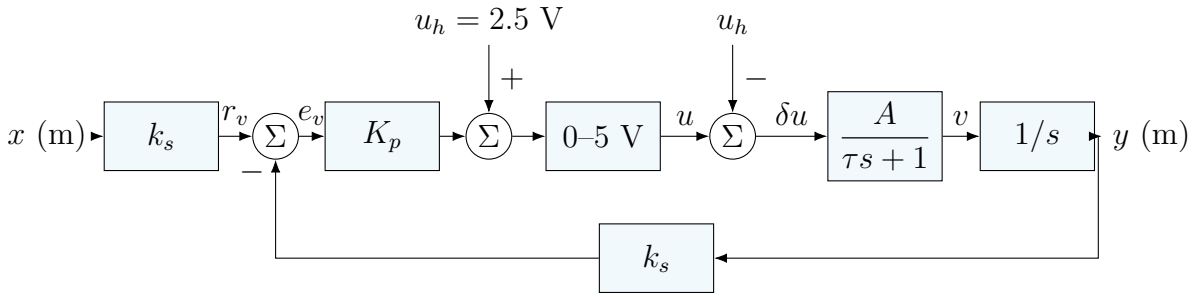


Figure 2.2: Complete voltage/altitude feedback loop. The plant accepts a deviation command after hover subtraction.

Figure 2.2 applies k_s once to each altitude signal. The hover terms cancel in the unsaturated deviation model; gravity must not be subtracted a second time. Although Session 2 mentions a 10 V supply restriction, the stricter 0–5 V plant-interface limit is used here.

2.2.3 Closed-Loop Representation

In the unsaturated region, $\Delta U = K_p k_s (X - Y)$. Substituting into Equation (2.5) and collecting the output terms gives

$$Y = G_y K_p k_s (X - Y) \;\Rightarrow\; (1 + G_y K_p k_s) Y = G_y K_p k_s X. \tag{2.7}$$

Define $q = AK_p k_s$ and $D(s) = \tau s^2 + s + q$. The reference and position-error transfer functions are

$$L(s) = K_p k_s \frac{A}{s(\tau s + 1)}, \tag{2.8}$$

$$\frac{Y(s)}{X(s)} = \frac{L(s)}{1 + L(s)} = \frac{q}{D(s)}, \tag{2.9}$$

$$\frac{E_y(s)}{X(s)} = 1 - \frac{Y(s)}{X(s)} = \frac{\tau s^2 + s}{D(s)}, \qquad E_y = X - Y. \tag{2.10}$$

An additional acceleration enters Equation (2.2) directly. Therefore

$$\frac{Y(s)}{D_a(s)} = \frac{\tau}{D(s)}, \qquad \frac{Y(s)}{G(s)} = -\frac{\tau}{D(s)} \quad \text{for uncompensated gravity.} \tag{2.11}$$

The 2.5 V bias compensates nominal gravity in the actual implementation.

2.2.4 Steady-State Properties

The loop is Type 1 because $L(s)$ contains one pole at the origin. For $K_p > 0$, the characteristic polynomial $s^2 + bs + k_T K_p k_s$ has positive coefficients and is stable. The final-value theorem gives zero step-reference error. For a ramp $x(t) = a_r t$, $K_v = \lim_{s \to 0} sL(s) = AK_p k_s$, so $e_\infty = a_r / K_v$ is finite; a parabolic reference has infinite steady-state error.

A constant disturbance behaves differently from a reference step. Applying the final-value theorem to Equation (2.11) gives

$$y_\infty - x_\infty = \frac{d_0}{k_T K_p k_s} = \frac{\delta u_b}{K_p k_s} \quad \text{when } d_0 = k_T \delta u_b. \tag{2.12}$$

For a 5 m reference and $K_p = 0.01432$, the 8% error allowance requires

$$|\delta u_b| < 0.08(5)(0.01432)(0.93) = 0.005327\,\text{V}. \tag{2.13}$$

Hover bias must consequently be calibrated to about 5.3 mV. The brief does not specify disturbance amplitude or injection point; disturbance compliance is evaluated for explicitly defined cases in Appendix E.

3 System Identification

3.1 Introduction

Identification removes the position integrator by differentiating the metre-valued altitude. The resulting first-order velocity response permits independent estimates of DC gain and time constant. The sensor conversion is fitted separately from unsaturated voltage/altitude pairs.

3.2 Methodology

3.2.1 Step-Testing

The Session 1 input circuit described in the brief uses a potentiometer to set the ADC command. The retained record starts at rest near 2.5 V and changes to 5 V. The nominal input increment is $\Delta u = 5.0 - 2.5 = 2.5$ V. The final voltage is held long enough to identify a steady velocity. Figure 3.1 shows the input, position and sensor channels from this run.

3.3 Experimental Setup and Data Files

The Session 1 data folder contains the unmodified `PTRJUN003_raw.csv`; Appendix G gives its provenance. Of 1,634 rows, the first 31 precede a clock reset from 9.5 to 0.1 s and contain zero altitude. They are excluded from analysis. The retained 1,603 samples span 0.1–160.3 s at 0.1 s spacing. The reset is not another trial.

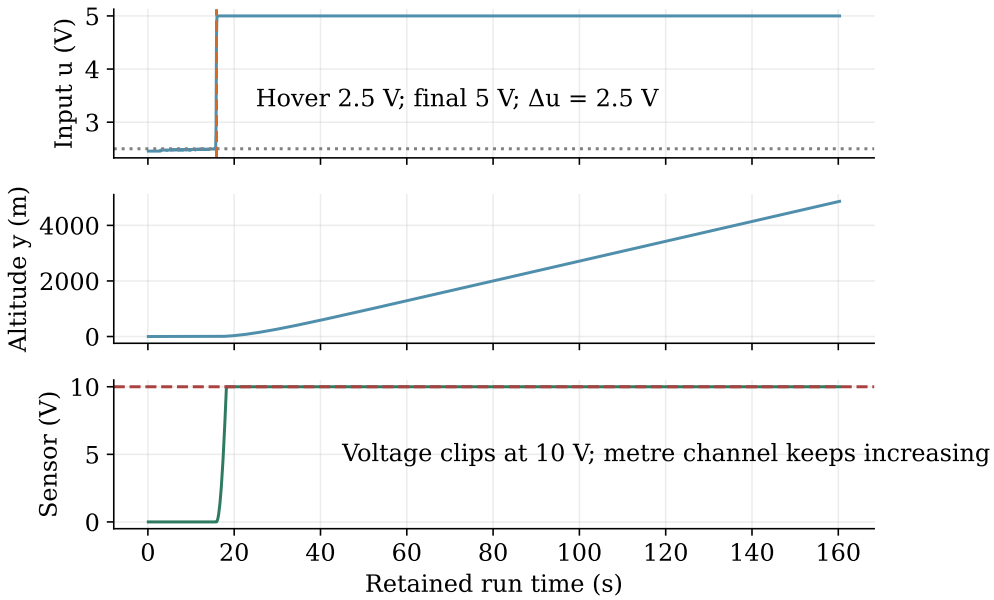


Figure 3.1: Recorded identification input and outputs. The sensor-voltage plateau is clipping at 10 V; altitude continues increasing.

3.4 Data Processing

3.4.1 Velocity Computation

The metre channel is differentiated using a backward difference:

$$v_i = \frac{y_i - y_{i-1}}{t_i - t_{i-1}}, \qquad t_{v,i} = \frac{t_i + t_{i-1}}{2}. \tag{3.1}$$

For example, the 16.9 and 17.0 s positions are 2.256 and 2.703 m, giving $v = (2.703 - 2.256)/0.1 = 4.47$ m/s at midpoint 16.95 s. No smoothing is applied. The stored speed column agrees to CSV rounding. Differentiating the clipped voltage channel would incorrectly produce zero velocity; the metre channel avoids this error.

3.4.2 Estimation of Gain and Time Constant

The 404 velocity samples with endpoint time $t \geq 120$ s give $v_\infty = 35.68748$ m/s. With $v_0 = 0$,

$$A = \frac{v_\infty - v_0}{\Delta u} = \frac{35.68748 - 0}{5.0 - 2.5} = 14.27499\,\mathrm{m/(s\,V)}. \tag{3.2}$$

The first-order step response is $v(t) = v_\infty[1 - e^{-(t-t_0)/\tau}]$. At one time constant, $1 - e^{-1} \approx 0.632$, so $v_{63} = 0.632(35.68748) = 22.55448$ m/s. The measured crossing lies between 24.0 and 24.1 s:

$$t_{63} = 24.0 + \frac{22.55448 - 22.53}{22.69 - 22.53}(24.1 - 24.0) = 24.01530\,\mathrm{s}, \tag{3.3}$$

$$\tau_{63} = t_{63} - t_0 \approx 24.01530 - 15.9 = 8.11530\,\mathrm{s}. \tag{3.4}$$

A least-squares exponential fit over 17–60 s gives $V_\infty = 35.68747$ m/s, $\tau = 8.10393$ s and $t_0 = 15.91584$ s. Endpoint-labelled differences are used in that fit; the midpoint shift is absorbed in t_0, leaving τ unchanged.

3.4.3 Sensor Calibration (V/m)

Using 22 nonzero pairs with $0 < y_v < 9.9$ V, the zero-intercept least-squares coefficient is

$$k_s = \frac{\sum_i y_i y_{v,i}}{\sum_i y_i^2} = 0.9300026\,\mathrm{V/m} \approx 0.93\,\mathrm{V/m}. \tag{3.5}$$

The single-point check at 17.0 s is $2.513/2.703 = 0.92971$ V/m. The regression uses multiple unsaturated pairs rather than the clipped 10 V tail. Full processing equations and selected sample calculations are in Appendix E.

3.5 Results and Analysis

3.5.1 Step Response Analysis

Figure 3.2 compares the differentiated measurements with the exponential fit and shows the 63.2% crossing. Velocity tends to a constant while altitude continues rising, as expected for the lag-plus-integrator model.

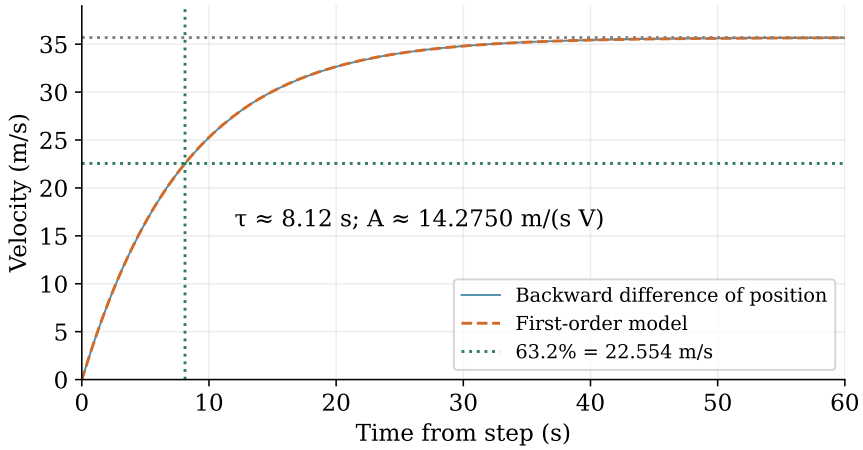


Figure 3.2: Velocity identification. The steady tail determines gain; interpolation and fitting determine the time constant.

3.5.2 Identification Results

Table 3.1 compares the two estimation methods. These are estimates from one run, not independent repeated trials. The rounded parameters reproduce the values used in the supplied Simulink model.

Table 3.1: Identification estimates and adopted design parameters.

Method	A [m/(s V)]	τ [s]	k_s [V/m]
Tail / 63.2% crossing	14.27499	8.11530	—
Exponential / sensor fit	14.27499	8.10393	0.930003
Adopted model	14.2748	8.1000	0.9300

3.6 Transfer Function Approximation

Substitution into Equation (2.4) and Equation (2.5) gives

$$\boxed{G_v(s) = \frac{14.2748}{8.1s + 1}, \qquad G_y(s) = \frac{14.2748}{s(8.1s + 1)}}. \tag{3.6}$$

The physical coefficients follow directly: $b = 1/8.1 = 0.1234568$ s^{-1} and $k_T = A/\tau = 14.2748/8.1 = 1.762321$ m/(s^2 V). The differences between fitted and rounded values are small relative to the specified 10% aerodynamic uncertainty.

3.7 Model Validation

The rounded model is evaluated over both the fitting interval (17–60 s) and a held-out tail ($t > 60$ s). It predicts

$$v_{\mathrm{mod}}(t) = 2.5A(1 - e^{-r/\tau}), \tag{3.7}$$

$$y_{\mathrm{mod}}(t) = 2.5A\{r - \tau(1 - e^{-r/\tau})\}, \qquad r = \max(t - t_0, 0). \tag{3.8}$$

The effective onset fitted above is retained. Figure 3.3 checks sensor linearity and velocity residuals; Table 3.2 reports error magnitudes.

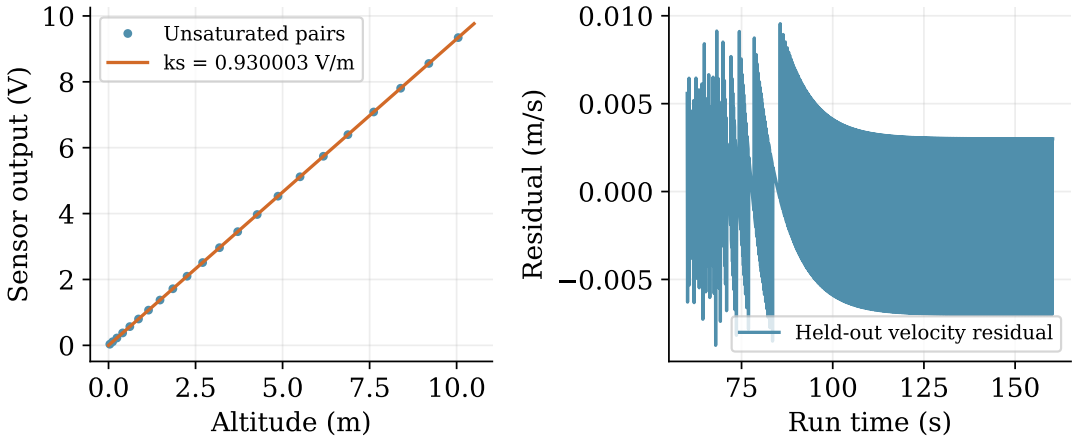


Figure 3.3: Sensor calibration from unsaturated pairs and velocity residuals outside the fitting interval.

Table 3.2: Validation errors using the rounded design model.

Comparison	Interval [s]	RMSE
Velocity in fitting window	17–60	0.00533 m/s
Held-out velocity	> 60	0.00436 m/s
Held-out position	> 60	1.686 m

RMSE is $\sqrt{N^{-1}\sum_i (z_i - z_{\mathrm{mod},i})^2}$. Relative to final velocity, the held-out error is 0.0122%. The position comparison in Figure F.1 reaches almost 4,865 m; small rounding and timing differences accumulate through integration, so its absolute RMSE is larger.

The residuals support the first-order velocity structure and the adopted parameter values for controller design. This validation tests another interval of the same run. It does not establish repeatability across independent experiments or the accuracy of a model-check submission to the laboratory system.

4 Controller Design

4.1 Design Process

4.1.1 Plant for Design

The design uses Equation (3.6), $k_s = 0.93$ V/m and a 2.5 V hover bias. Its characteristic equation is

$$1 + K_p k_s G_y(s) = 0 \quad \Rightarrow \quad 8.1s^2 + s + 14.2748(0.93)K_p = 0. \tag{4.1}$$

4.1.2 Design Specifications

For a standard underdamped second-order response,

$$M_p = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \quad M_p < 30\% \Rightarrow \zeta > \frac{-\ln 0.3}{\sqrt{\pi^2 + \ln^2 0.3}} = 0.357857. \tag{4.2}$$

Matching the monic form of Equation (4.1) to $s^2 + 2\zeta\omega_n s + \omega_n^2$ yields

$$\omega_n = \sqrt{\frac{AK_p k_s}{\tau}}, \qquad \zeta = \frac{1}{2\sqrt{\tau A K_p k_s}}, \qquad \sigma = \zeta\omega_n = \frac{1}{2\tau}. \tag{4.3}$$

Therefore $K_p < 1/(4\tau A k_s \zeta_{\min}^2) = 0.0181544$ for nominal overshoot compliance. Type 1 tracking is ideal only after bias and reference/sensor scaling are calibrated. A practical nominal validation target is a 2% band within 60 s; for drag-only uncertainty, 75 s is reasonable. These are reporting targets added to the four official requirements, not guarantees for uncalibrated component corners.

4.1.3 Gain Selection (Root Locus)

The open-loop poles are 0 and $-1/\tau = -0.1234568$; there are no finite zeros. The real-axis locus lies between the poles, and its two asymptotes have angles $\pm 90°$ about centroid $-1/(2\tau)$. Differentiating $K_p(s) = -s(\tau s + 1)/(Ak_s)$ gives

$$\frac{dK_p}{ds} = 0 \Rightarrow s_b = -\frac{1}{2\tau} = -0.0617284, \quad K_{p,b} = \frac{1}{4\tau A k_s} = 0.00232489. \tag{4.4}$$

The supplied gain $K_p = 0.01432$ is above breakaway and below the nominal overshoot limit. Since the complex branches have fixed real part -0.0617284, a 30 s decay-envelope target ($\sigma \approx 4/30$) cannot be imposed by proportional gain alone. Detailed angle and magnitude checks are in Appendix E; the root-locus code folder reproduces the method.

4.1.4 Design Summary

Substituting $K_p = 0.01432$ gives $q = 14.2748(0.01432)(0.93) = 0.1901061$ and

$$G_{cl}(s) = \frac{0.1901061}{8.1s^2 + s + 0.1901061} = \frac{0.02346989}{s^2 + 0.1234568s + 0.02346989}. \tag{4.5}$$

The roots and response quantities in Table 4.1 are calculated from Equation (4.3) and checked against Figure 4.1.

Table 4.1: Manual nominal-controller calculations.

Quantity	Calculation	Result
Natural frequency	$\sqrt{0.1901061/8.1}$	0.153199 rad/s
Damping ratio	$1/[2\sqrt{8.1(0.1901061)}]$	0.402930
Damped frequency	$\omega_n\sqrt{1-\zeta^2}$	0.140212 rad/s
Closed-loop poles	$-\sigma \pm j\omega_d$	$-0.0617284 \pm j0.140212$
Overshoot	$100e^{-\pi(0.402930)/\sqrt{1-0.402930^2}}$	25.0804%
Peak time	$\pi/0.140212$	22.4060 s
Peak altitude	$5(1+0.250804)$	6.25402 m

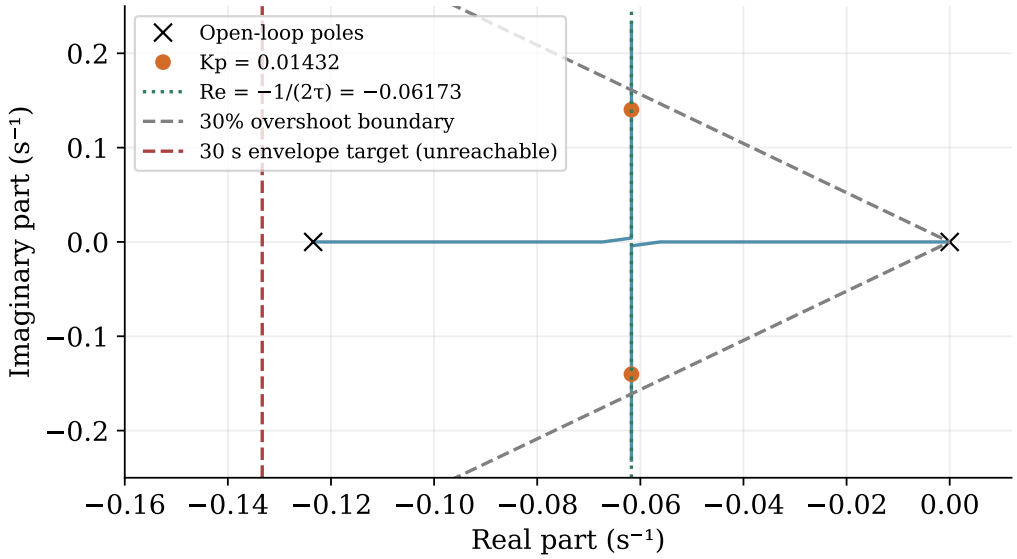


Figure 4.1: Proportional root locus. The selected poles satisfy the 30% overshoot boundary; the 30 s envelope target is beyond the attainable complex branches.

The exact step response is

$$y(t) = 5\left[1 - e^{-\sigma t}\left(\cos\omega_d t + \frac{\sigma}{\omega_d}\sin\omega_d t\right)\right]. \tag{4.6}$$

The usual $4/\sigma = 64.8$ s settling approximation is conservative here. The envelope bound $-\ln[0.02\sqrt{1-\zeta^2}]/\sigma = 64.81$ s and the actual final 2% crossing, 54.89 s, are different quantities.

4.2 Simulink Model Design

4.2.1 Block Diagram Implementation

The implementation follows Figure 2.2: reference scaling, error subtraction, K_p, hover addition, 0–5 V saturation, hover subtraction, the velocity lag and an integrator. The original model and scope are in Appendix C. The supplied scope includes startup from zero while the reference begins at 2.5 m. For an unambiguous demonstration test, the regenerated model uses zero initial altitude/velocity and a 0 to 5 m reference step.

4.2.2 Representative Simulation Results

Python evaluates Equation (4.6) on a 0.005 s grid. Figure 4.2 shows altitude and the corresponding command. The MATLAB folder contains independent verification and a Simulink model-building script; those scripts are supplied for execution in MATLAB.

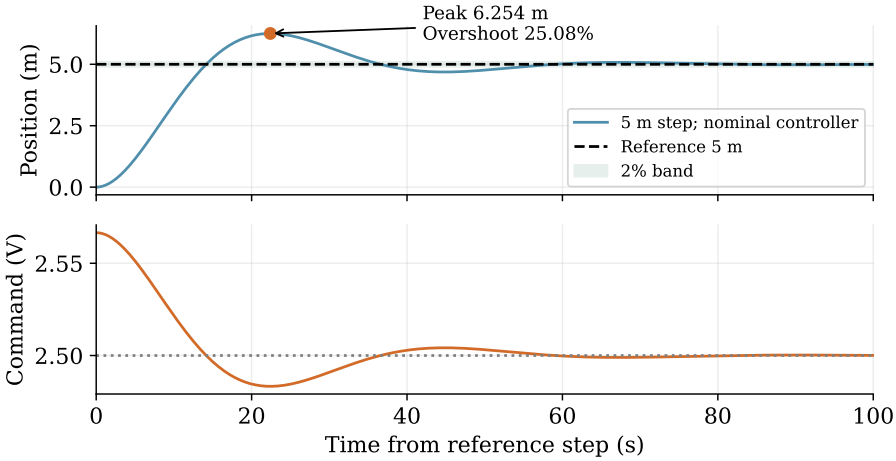


Figure 4.2: Nominal 5 m step prediction, with the 2% altitude band and actuator command.

The simulated peak is 6.25402 m at 22.405 s, agreeing with Table 4.1. The final reference-band entry is $T_{s,2\%} = 54.89$ s. Command limits are inactive: $u(0^+) = 2.5 + 0.01432(0.93)(5) = 2.56659$ V and $u_{\min} = 2.48330$ V. The nominal 60 s target is met.

For aerodynamic uncertainty, b varies while k_T remains fixed; $\tau = 1/b$ and $A = k_T/b$ therefore vary together. Table 4.2 and Figure F.2 show that drag uncertainty alone satisfies the 30% limit and proposed 75 s target.

Table 4.2: Predicted performance under aerodynamic uncertainty.

Drag	Peak [m]	Overshoot [%]	$T_{s,2\%}$ [s]
$0.9b_0$	6.473	29.45	70.80
b_0	6.254	25.08	54.89
$1.1b_0$	6.058	21.15	54.59

4.2.3 Controller Hardware Implementation

The controller uses a difference amplifier, a non-inverting amplifier and a biased output stage. The labelled circuit is in Figure D.1; the final Veroboard photograph is in Figure A.1. The supplied circuit files establish the design values in Table 4.3; the photograph alone cannot verify the underside wiring.

Table 4.3: Documented circuit components and nominal stage functions.

Stage	Values	Function
U1	$R_1 = R_3 = 100$ kΩ; $R_2 = R_4 = 1$ kΩ	$V_1 = 0.01(r_v - y_v)$; reference on +, sensor on −.
U2	$R_5 = 4.7$ kΩ; $R_6 = 10$ kΩ	$V_2 = (1 + R_5/R_6)V_1 = 1.47V_1$.
U3	R_7–$R_{10} = 100$ kΩ	Positive proportional correction plus adjustable 2.5 V bias.
Bias	$R_{11} = 3.9$ kΩ to ground; $R_{12} = 1.2$ kΩ across P1; $R_{13} = 22$ kΩ to −15 V; P1 = 10 kΩ	Loaded wiper feeds R_9; trim at equal reference and sensor inputs.
Supply	Three 741-type stages, ±15 V; six 100 nF capacitors in design	Common DAQ ground; one capacitor from each rail to ground at each op-amp.

The unloaded product $0.01(1.47) = 0.0147$ overestimates gain. At the calibrated setting $W = 0.783081$, the bias network has $V_{\text{th}} = -2.635575$ V and $R_{\text{th}} = 5.423001$ kΩ. Thus

$$u_0 = -\frac{100}{100 + 5.423001}(-2.635575) = 2.500000\,\text{V}, \tag{4.7}$$

$$K_{p,\text{circuit}} = 0.01(1.47)\frac{1 + 100/(100 + 5.423001)}{2} = 0.0143219. \tag{4.8}$$

This agrees with the simulation gain to 0.014%. Appendix D gives the complete loading derivation, rail connections and DAQ mapping.

Independent resistor variation also changes the reference/feedback ratio. Let $a_f = R_4/R_3$, $a_r = (1+a_f)R_2/(R_1+R_2)$, $g_2 = 1+R_5/R_6$ and $g_3 = (1+R_{10}/(R_9+R_{\text{th}}))R_8/(R_7+R_8)$. After hover recalibration,

$$K_f = a_f g_2 g_3, \qquad y_\infty/x_\infty = a_r/a_f. \tag{4.9}$$

The 2,048 loop-resistor/drag corners yield up to 48.79% tracking error and 38.95% overshoot (Figure E.1). Holding the bias network nominal makes this an optimistic subset, yet it already disproves full component robustness. Joint scalar uncertainty ($0.9b_0$, $1.1K_p$) gives 31.42% overshoot. Calculations and practical corrections follow in Appendix E and Section 5.3.

5 Controller Tests

5.1 Test Procedure

The documented DAQ mapping is DAC-1 for reference, DAC-0 for sensor feedback, ADC-1 for command and ADC-2 for sensor monitoring. Equal reference and sensor inputs are used to trim the zero-error command to 2.500 V. The retained test display then shows an altitude response from near zero to a reference of 5.033 m. The display, including its student identifier and numerical readouts, is preserved in Appendix B.

5.2 Experimental Results

At 86.10 s, the recorded reference is 5.033 m and height is 4.897 m. The display reports 0.137 m tracking error, 2.498 V command and 4.553 V sensor output. The computed error $5.033 - 4.897 = 0.136$ m differs from the tracking readout by only 0.001 m. The sensor check $4.553/4.896 = 0.92994$ V/m agrees with Session 1.

The blue response in Figure 5.1 is digitised from the original screenshot using its 0–120 s and 0–20 m graph axes. Resolution is 0.148 s/pixel and 0.0417 m/pixel. It gives an approximate peak of 6.229 m at 22.02 s. This plot is an estimate from image evidence, not an additional logged trial.

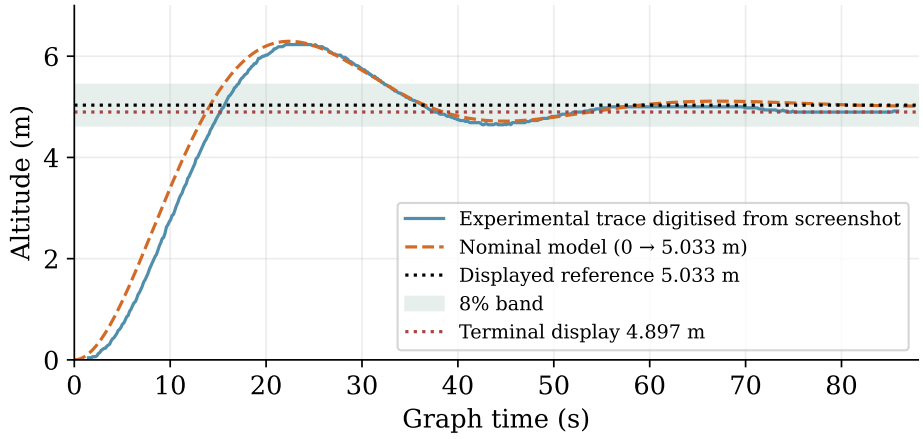


Figure 5.1: Retained experimental altitude trace compared with the nominal prediction for the same 5.033 m reference.

The calculated experimental metrics are

$$\eta_{\text{track}} = 100\left(1 - \frac{|5.033 - 4.897|}{5.033}\right) = 97.30\%, \tag{5.1}$$

$$M_{p,\text{exp}} \approx 100\frac{6.22917 - 5.033}{5.033 - 0} = 23.77\%. \tag{5.2}$$

This overshoot uses the commanded step. Peak uncertainty is about 0.83 percentage points per pixel. Appendix E also checks overshoot relative to the terminal height.

5.3 Performance Analysis

Table 5.1 compares the retained test and uncertainty predictions with the four laboratory requirements.

Table 5.1: Controller evaluation against the laboratory requirements.

Criterion	Evidence	Evaluation
Tracking $> 92\%$	Accuracy 97.30%; error 0.136 m.	Pass: retained nominal test.
Overshoot $< 30\%$	Model 25.08%; experiment 23.77% about reference, 27.20% about terminal height.	Pass: nominal responses.
Aerodynamics $\pm 10\%$	Predicted overshoot 21.15–29.45%.	Pass: drag-only model cases.
Components $\pm 10\%$	Corner error 48.79%; overshoot 38.95%.	Fail: independent tolerances.

5.4 Discussion

The final display is inside the 8% reference band but outside the 2% band. Nominal settling is 54.89 s; experimental 2% settling is unconfirmed. Joint drag/gain overshoot is 31.42%. A 2 mV bias predicts 0.150 m displacement (Figure F.3).

6 Conclusion and Recommendations

6.1 Conclusion

Identification supports $G_y(s) = 14.2748/[s(8.1s + 1)]$ and $k_s = 0.93$ V/m. Bias loading gives $K_p \approx 0.01432$. Nominal tracking and overshoot pass; component mismatch and joint uncertainty prevent full robustness.

6.2 Recommendations

1. Match/trim U1 ratios to below 1% mismatch and verify both signal slopes.
2. Use matched 680 Ω U1 resistors: proposed gain 0.00974, joint overshoot 23.44%.
3. Recalibrate to 2.500 V; the revised 5 m bias allowance is 3.623 mV.
4. Retain repeated raw steps and defined disturbance tests to assess robustness.

6.3 Final Remarks

Calculations are in Appendix E; source files are linked in Appendix H. Proposed changes require retesting.

References

[1] University of Cape Town, Department of Electrical Engineering, *EEE3094S Lab 2: Proportional Control*, 2026. Supplied laboratory brief and rubrics.

[2] UCT, *EEE3094S Lab 2/3 LaTeX Example*, supplied template. N. Mathebula, *Lab 2 Report*, 2025, supplied structure example.

[3] MathWorks, *stepinfo*. Online documentation.

[4] MathWorks, *rlocus*. Online documentation.

A Photographs of the Final Controller

Figure A.1 documents the soldered Veroboard implementation described in Section 4.2.3. The board has three eight-pin op-amp packages and a blue bias trimmer. The photograph records the final physical assembly; it does not establish every underside connection or measured resistance.

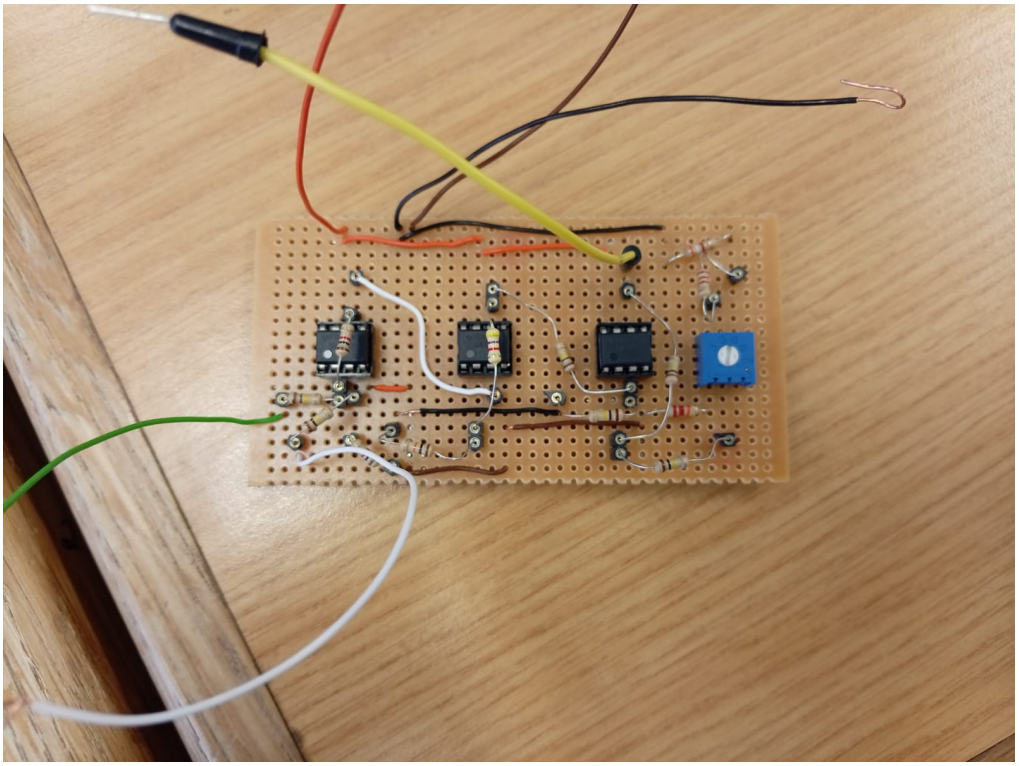


Figure A.1: Supplied final Veroboard photograph. The three op-amp stages and adjustable bias trimmer are visible.

The circuit design specifies 741-type devices and bipolar supply rails. Device markings and decoupling connections should be checked against the physical board when verifying the as-built netlist. No photographs of handwritten calculations are used in this report.

B Experimental Demonstration Record

The original display in Figure B.1 supports the response evaluated in Section 5. The red trace is the reference and the blue trace is altitude. This image is also retained in `data/session2/hardware_step_test.jpeg`.

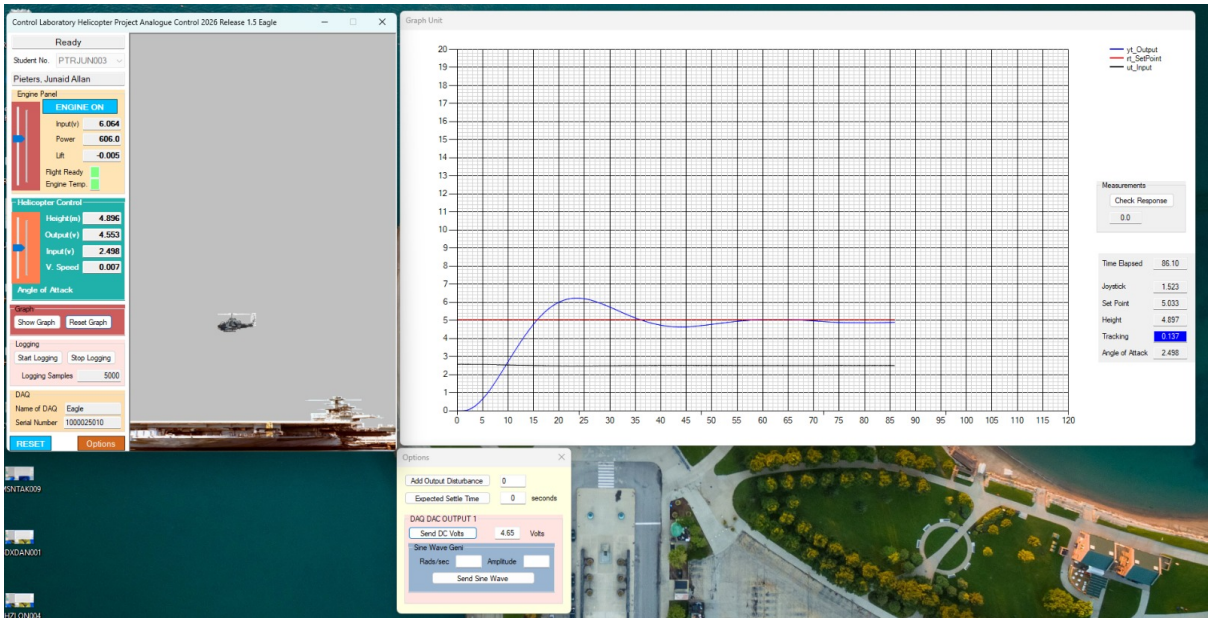


Figure B.1: Original controller-test display for PTRJUN003, showing the altitude response and terminal readouts.

The numerical values transcribed in Table B.1 support the tracking and sensor checks.

Table B.1: Numerical readouts transcribed from the retained test display.

Field	Displayed value
Student number	PTRJUN003
Set point	5.033 m
Height	4.897 m
Tracking error	0.137 m
Angle of attack / command	2.498 V
Height (left panel)	4.896 m
Sensor output (left panel)	4.553 V
Elapsed time	86.10 s

The 0.001 m differences between the readouts are consistent with rounding or asynchronous display updates. The assessor's marking sheet and signature were not supplied, so an awarded demonstration mark is not claimed.

C Original Simulink Evidence

Figure C.1 preserves the supplied model image. Its visible values match the identified plant, sensor coefficient and selected gain used in Section 4.

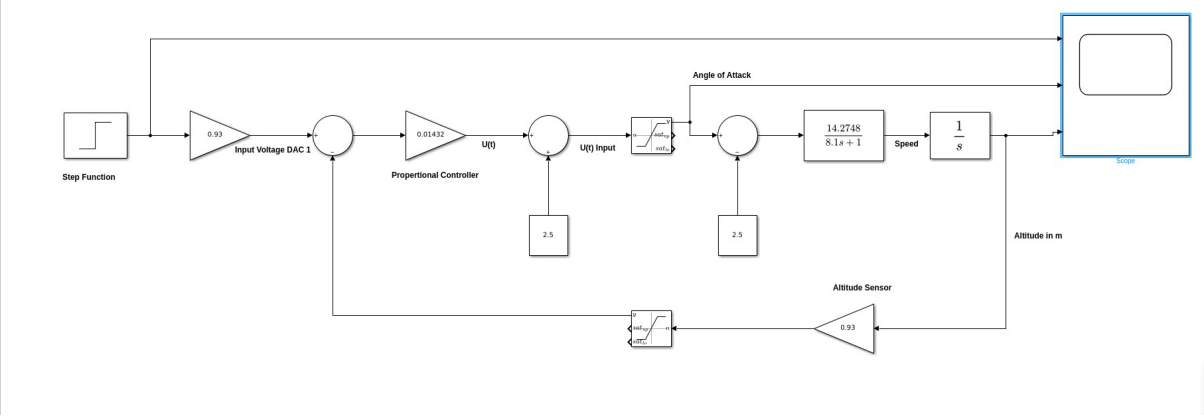


Figure C.1: Supplied Simulink model: $k_s = 0.93$, $K_p = 0.01432$, $u_h = 2.5$ V, velocity plant $14.2748/(8.1s + 1)$ and position integrator.

Figure C.2 begins at a 2.5 m reference while the output starts near zero; the reference subsequently changes to 5 m at approximately 2.5 s. The trace therefore combines startup and a reference change.

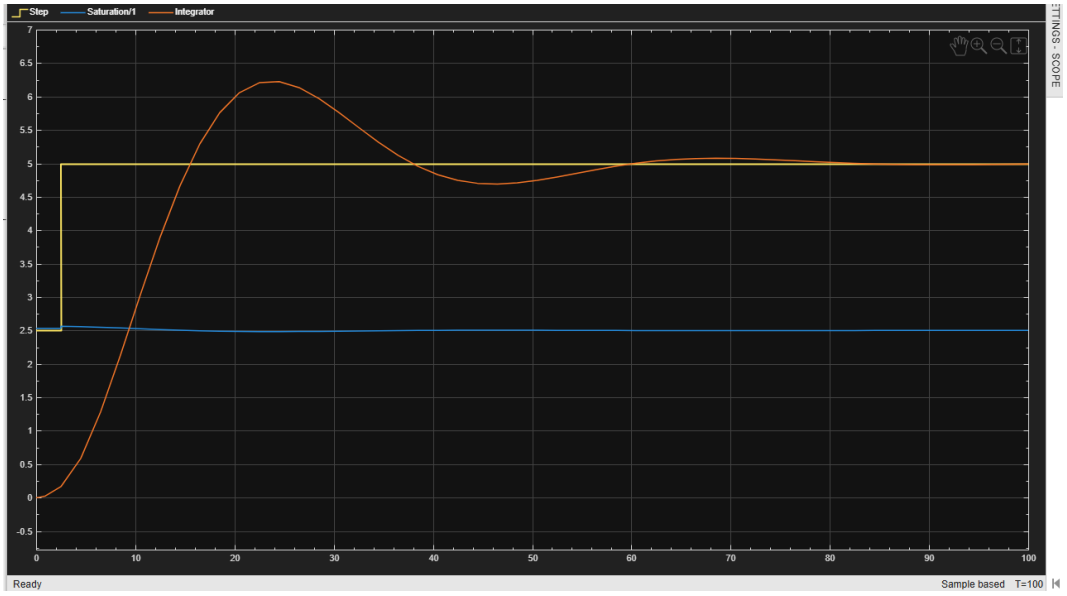


Figure C.2: Original supplied scope with reference and altitude traces. The initial output is not settled at the initial reference.

The regenerated 0 to 5 m step in Figure 4.2 avoids combining these initial conditions. The included `matlab/build_simulink_model.m` creates a native model with explicit 0–5 V command saturation when executed in Simulink. The original native `.slx` file was not supplied.

D Complete Circuit and Interface

D.1 Signal Stages

Figure D.1 uses the component numbering of the supplied circuit files. Its bias input connects to the P1 wiper in Figure D.2; the two figures together define the complete controller signal and bias networks.

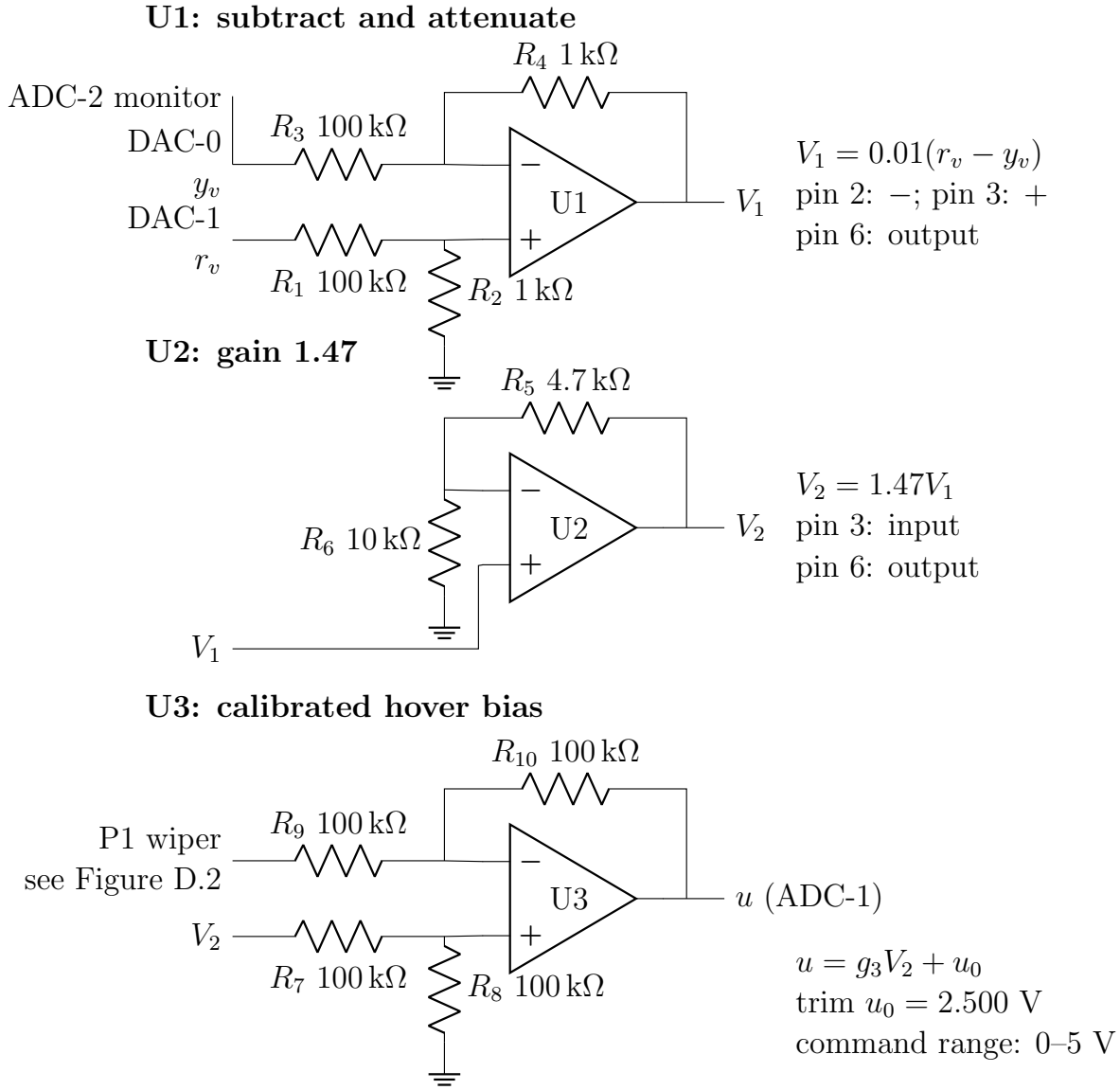


Figure D.1: Labelled three-stage proportional controller. DAC-0 feeds the inverting sensor branch; DAC-1 feeds the non-inverting reference branch. ADC-2 monitors the sensor and ADC-1 receives the command.

All three op-amps use pin 7 for +15 V, pin 4 for −15 V, pin 2 for the inverting input, pin 3 for the non-inverting input and pin 6 for output. Pins 1 and 5 (offset null) and pin 8 (NC) are unused in the documented 741 design. Table D.1 gives the interface mapping.

D.2 Bias Network and Supply Decoupling

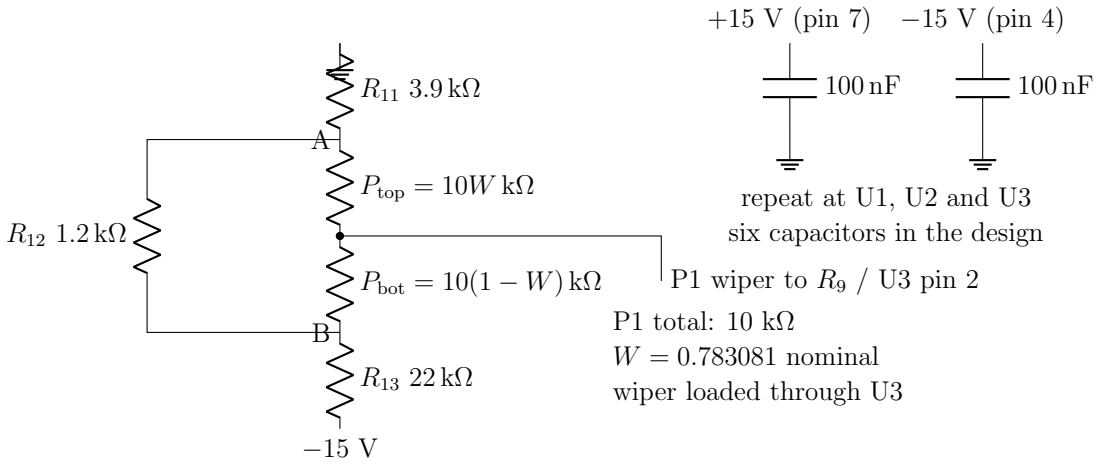


Figure D.2: Bias potentiometer represented by its two segments, shunt resistor and end resistors; supply decoupling is repeated at each op-amp.

Table D.1: DAQ and supply connections.

Connection	Controller node
DAC-1	r_v, U1 reference branch through R_1.
DAC-0	y_v, U1 sensor branch through R_3.
ADC-2	Sensor monitor at y_v.
ADC-1	Command u, U3 pin 6; permitted plant range 0–5 V.
Ground	Shared by DAQ, bias network and bench supply.
±15 V	Op-amp rails; these do not define the permitted plant command.

The LTspice source and supplied circuit image place R_{11} at ground. The supplied KiCad file labelled that endpoint +15 V; its bundled copy is corrected to ground to match those two records. This correction is documented in `hardware/design_basis.md`. Calibration sets $r_v = y_v$, then adjusts P1 until the loaded ADC-1 command is 2.500 V. Keep the wiper fixed during testing.

D.3 Bias Loading and Stage Gain Calculations

With the wiper unloaded, $R_P = P1 \parallel R_{12} = 10 \parallel 1.2 = 1.0714286$ kΩ. The end-to-end current and pot-end voltages are

$$I = \frac{-15}{3.9 + 1.0714286 + 22} = -0.556143\,\text{mA}, \tag{D.1}$$

$$V_A = I(3.9) = -2.168962\,\text{V}, \qquad V_B = I(3.9 + 1.0714286) = -2.764831\,\text{V}, \tag{D.2}$$

$$V_{\text{th}} = V_A + W(V_B - V_A). \tag{D.3}$$

Set the supply sources to zero and inject a 1 A test current at the wiper. The node-conductance equations for A, B and the wiper are

$$\begin{bmatrix} G_{11} + G_{12} + G_t & -G_{12} & -G_t \\ -G_{12} & G_{13} + G_{12} + G_b & -G_b \\ -G_t & -G_b & G_t + G_b \end{bmatrix} \begin{bmatrix} V'_A \\ V'_B \\ V'_w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \tag{D.4}$$

where $G_i = 1/R_i$, $G_t = 1/(WP1)$ and $G_b = 1/[(1-W)P1]$. The test voltage V'_w gives R_{th}. Solving the hover equation

$$-\frac{R_{10}}{R_9 + R_{\text{th}}(W)} V_{\text{th}}(W) = 2.5 \tag{D.5}$$

gives $W = 0.783080594$, $R_{\text{th}} = 5423.0014\ \Omega$ and $V_{\text{th}} = -2.6355750$ V. The loaded zero-error wiper is -2.500 V.

For U3, define $r = R_{10}/(R_9 + R_{\text{th}}) = 0.94855960$. Since the non-inverting divider gives $V_+ = V_2/2$, virtual short and KCL give

$$u = (1 + r)V_+ - rV_{\text{th}} = \frac{1 + r}{2} V_2 - rV_{\text{th}}, \tag{D.6}$$

$$g_3 = \frac{1 + 0.94855960}{2} = 0.97427980, \tag{D.7}$$

$$K_p = \left(\frac{1}{100}\right)\left(1 + \frac{4.7}{10}\right)(0.97427980) = 0.014321913. \tag{D.8}$$

The ideal buffered result 0.0147 differs by 2.57%, while the selected model value 0.01432 differs from the loaded calculation by only 0.0134%. These are circuit-model calculations; a measured input/output slope of the final board was not supplied.

E Detailed Calculations

E.1 Identification and Timing

Table E.1 gives selected original samples and derived velocities. The finite manual step is evident in the input column. Endpoint times are listed for consistency with the CSV; each derivative physically belongs to its interval midpoint.

Table E.1: Selected identification samples and backward-difference results.

t [s]	u [V]	y_v [V]	y [m]	v_i [m/s]
15.7	2.495	0.000	0.000	0.00
15.8	2.908	0.000	0.000	0.00
15.9	4.607	0.000	0.000	0.00
16.0	5.000	0.034	0.037	0.37
16.9	5.000	2.098	2.256	4.08
17.0	5.000	2.513	2.703	4.47
23.9	5.000	10.000	104.823	22.36
24.0	5.000	10.000	107.076	22.53
24.1	5.000	10.000	109.345	22.69

$$v(24.0) = (107.076 - 104.823)/0.1 = 22.53\,\mathrm{m/s}, \tag{E.1}$$

$$v(24.1) = (109.345 - 107.076)/0.1 = 22.69\,\mathrm{m/s}, \tag{E.2}$$

$$\bar{v}_\infty = \frac{1}{404}\sum_{t_i \geq 120} \frac{y_i - y_{i-1}}{0.1} = 35.68747525\,\mathrm{m/s}, \tag{E.3}$$

$$A = (35.68747525 - 0)/2.5 = 14.27499010\,\mathrm{m/(s\,V)}, \tag{E.4}$$

$$v_{63} = 0.632(35.68747525) = 22.55448436\,\mathrm{m/s}, \tag{E.5}$$

$$t_{63} = 24.0 + 0.1\frac{22.55448436 - 22.53}{22.69 - 22.53} = 24.01530272\,\mathrm{s}, \tag{E.6}$$

$$\tau_{63} = 24.01530272 - 15.9 = 8.11530272\,\mathrm{s}. \tag{E.7}$$

The exponential fit minimises $\sum_{17 \leq t_i \leq 60}[v_i - V_\infty(1 - e^{-(t_i - t_0)/\tau})]^2$. Its outputs are $V_\infty = 35.68747059$ m/s, $\tau = 8.10393362$ s and $t_0 = 15.91583783$ s. These two methods corroborate rounding to $\tau = 8.1$ s. They do not create two experimental trials.

The sensor least-squares fit in Equation (3.5) excludes zero and clipped pairs. The check $k_s(2.703) = 0.93(2.703) = 2.51379$ V agrees with the recorded 2.513 V within rounding.

E.2 Root-Locus Angle and Magnitude Checks

The locus for variable K_p uses $G_0(s) = k_s G_y(s) = (Ak_s/\tau)/[s(s + 1/\tau)]$, where $Ak_s/\tau = 1.638959\ \mathrm{s}^{-2}$. At the selected upper pole $s_d = -0.0617284 + j0.1402123$,

$$\angle s_d = 113.76147^\circ, \qquad \angle(s_d + 1/\tau) = 66.23853^\circ, \tag{E.8}$$

$$\angle G_0(s_d) = -113.76147^\circ - 66.23853^\circ = -180^\circ, \tag{E.9}$$

$$K_p = \frac{1}{|G_0(s_d)|} = \frac{|s_d|\,|s_d + 1/\tau|}{Ak_s/\tau} = \frac{(0.153198845)^2}{1.63895852} = 0.01432. \tag{E.10}$$

Both the negative-feedback angle criterion and magnitude criterion are satisfied. The quadratic roots independently give the same pole pair.

E.3 Nominal Step and Settling Calculation

Substituting into Equation (4.6),

$$y(t) = 5\left[1 - e^{-0.061728395t}\left(\cos(0.140212308t) + 0.44025\sin(0.140212308t)\right)\right]. \tag{E.11}$$

The first peak occurs at $t_p = \pi/\omega_d = 22.405969$ s. Since $\cos\pi = -1$ and $\sin\pi = 0$,

$$y(t_p) = 5[1 + e^{-0.061728395(22.405969)}] = 6.254019\,\mathrm{m}, \tag{E.12}$$

so $M_p = (6.254019 - 5)/5 \times 100 = 25.080375\%$.

The error envelope satisfies $|y - 5|/5 \le e^{-\sigma t}/\sqrt{1 - \zeta^2}$. Solving this bound for 2% gives

$$T_{\mathrm{env}} = -\frac{\ln(0.02\sqrt{1 - 0.4029299^2})}{0.061728395} = 64.80975\,\mathrm{s}. \tag{E.13}$$

Actual settling is found by evaluating the full response, locating the last sample for which $|y - 5| > 0.1$ m, and taking the next sample. A 0.005 s grid gives 54.89 s. This definition agrees with reference-based settling calculations [3].

Command headroom follows from $u = 2.5 + K_p k_s(5 - y)$: the highest command is 2.566588 V at the step, and the lowest is $2.5 - 0.01432(0.93)(1.254019) = 2.483299$ V at the altitude peak. Neither reaches the 0 or 5 V limit.

E.4 Aerodynamic and Component Uncertainty

With fixed k_T, the uncertain characteristic polynomial is $s^2 + b's + k_T K_p k_s$. The worst scalar case has $b' = 0.9b_0$ and $K_p' = 1.1K_{p,0}$. Applying the required damping ratio gives

$$K_{p,0} < \frac{(0.9b_0)^2}{4k_T k_s \zeta_{\min}^2(1.1)} = \frac{0.81}{1.1}(0.01815443) = 0.01336826. \tag{E.14}$$

At the supplied gain the worst-case damping is $(0.9/\sqrt{1.1})(0.4029299) = 0.34575$, and Equation (4.2) gives 31.42% overshoot. This explains why passing drag-only uncertainty is insufficient for the joint case.

For independent loop-resistor tolerances, use the U1 coefficients and U2/U3 gains defined in Section 4.2.3. Assign each of R_1–R_{10} either 0.9 or 1.1 times its nominal value, then use both drag extremes. There are $2^{10} \times 2 = 2048$ cases. The bias network is held at its nominal Thevenin resistance and zero-error output is recalibrated in each case. Table E.2 reports this optimistic subset.

Table E.2: Independent loop-resistor corner predictions.

Quantity	Range / worst value
Feedback gain K_f	0.009072–0.022784
Reference/feedback ratio a_r/a_f	0.672104–1.487864
Absolute DC tracking error	48.7864%
Overshoot about each case's own final altitude	38.9461%

For example, $R_1 = 0.9(100\,\text{k}\Omega)$, $R_2 = 1.1(1\,\text{k}\Omega)$, $R_3 = 1.1(100\,\text{k}\Omega)$ and $R_4 = 0.9(1\,\text{k}\Omega)$ give

$$a_f = 0.9/110 = 0.00818182, \quad \frac{a_r}{a_f} = \frac{(1 + 0.00818182)(1.1)/(90 + 1.1)}{0.00818182} = 1.487864. \tag{E.15}$$

Thus a 5 m reference tends to 7.43932 m even after hover recalibration. The type number cannot remove unequal reference and feedback scaling.

E.5 Practical Gain and Ratio Correction

The corner distributions in Figure E.1 separate dynamic overshoot from DC ratio error.

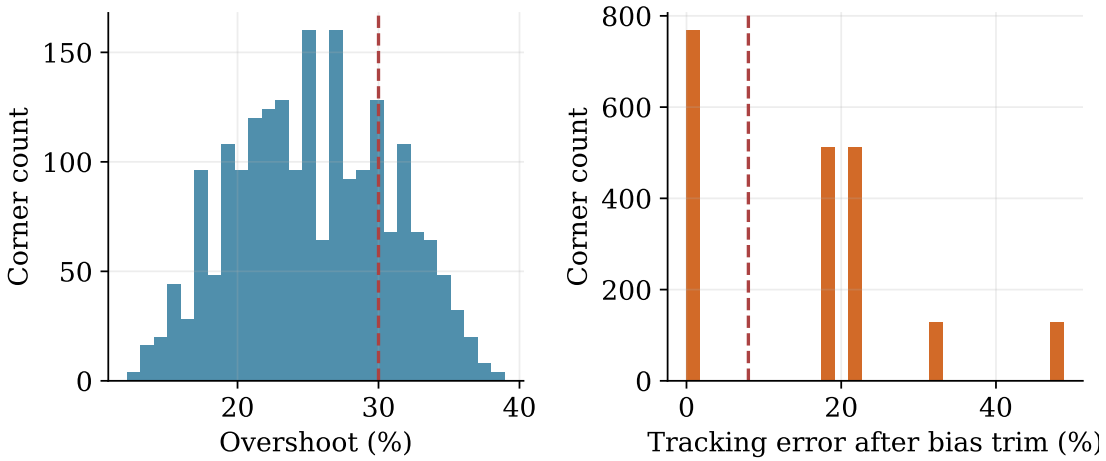


Figure E.1: Loop-resistor corners. Dashed limits are 30% overshoot and 8% tracking error. The histograms show corner counts, not probabilities.

Failure in this subset rejects a guarantee for all components. Bias-network and op-amp variations can add error. After matching the U1 ratios, a practical gain correction is to replace both 1 kΩ resistors with matched 680 Ω values:

$$K_{p,\mathrm{new}} = \frac{680}{100000}(1.47)(0.97427980) = 0.009738901, \tag{E.16}$$

$$\zeta_{\mathrm{worst}} = \frac{0.9b_0}{2\sqrt{k_T(1.1K_{p,\mathrm{new}})k_s}} \approx 0.4193, \tag{E.17}$$

$$M_{p,\mathrm{worst}} \approx 23.44\%, \tag{E.18}$$

$$|\delta u_b|_{\max} = 0.08(5)(0.009738901)(0.93) = 3.623\,\mathrm{mV}. \tag{E.19}$$

This scalar check assumes matched reference and feedback ratios; it is not a guarantee for independent resistor corners. Lower gain increases damping but tightens the hover-bias allowance. The board must be recalibrated and retested after any change.

E.6 Experimental Metrics and Uncertainty

The digitised plot uses pixel bounds $x = 602 \ldots 1414$ for 0–120 s and $p_y = 60 \ldots 540$ for 20–0 m. Therefore

$$t = \frac{x - 602}{1414 - 602}(120), \qquad h = \frac{540 - p_y}{540 - 60}(20), \tag{E.20}$$

$$\Delta t = 0.147783\,\text{s/pixel}, \qquad \Delta h = 0.041667\,\text{m/pixel}. \tag{E.21}$$

The extracted peak is 6.229167 m at 22.019704 s. With the displayed reference, terminal height and zero starting altitude,

$$e_f = 5.033 - 4.897 = 0.136\,\text{m}, \tag{E.22}$$

$$\eta_{\text{track}} = 100(1 - 0.136/5.033) = 97.297834\%, \tag{E.23}$$

$$M_{p,\text{ref}} = 100(6.229167 - 5.033)/5.033 = 23.766475\%, \tag{E.24}$$

$$\delta M_{p,\text{ref}} \approx 100(0.041667)/5.033 = 0.82787 \text{ percentage points}. \tag{E.25}$$

This practical overshoot metric uses the commanded reference. If the terminal display is used as a proxy for the actual settled output, the conventional final-output-normalised overshoot is $(6.229167 - 4.897)/4.897 \times 100 \approx 27.20\%$. Both estimates are below 30%, but the screenshot alone does not establish an exact asymptotic output. Table E.3 compares the analytical, simulated and experimental metrics.

Table E.3: Comparison of calculation, simulation and retained experiment.

Metric	Analytical	Simulation	Experiment
Reference	5.000 m	5.000 m	5.033 m
Peak	6.25402 m	6.25402 m	≈ 6.229 m
Peak time	22.40597 s	22.405 s	≈ 22.020 s
Reference overshoot	25.0804%	25.0804%	$\approx 23.7665\%$
Tracking accuracy	100% ideal	$\to 100\%$	97.2978% displayed
2% reference settling	64.81 s envelope	54.89 s	Not established

For the 5.033 m reference the 8% band is $5.033(1 \pm 0.08) = 4.63036$–$5.43564$ m. The 2% band is 4.93234–5.13366 m. The terminal height lies inside the first but outside the second. The discrepancy cannot be removed by redefining the settling band around the biased terminal value while claiming reference tracking.

The command-bias calculation uses Equation (2.12): $\delta y = 0.002/(0.01432 \times 0.93) = 0.150177$ m. The illustrative test in Figure F.3 injects a positive 2 mV bias at 70 s; it is a model prediction rather than a recorded disturbance experiment.

F Additional Validation and Response Plots

Figure F.1 checks the integrated position response; Figure F.2 compares the three aerodynamic cases listed in Table 4.2. The underlying values are retained as CSV and JSON in the `results` folder.

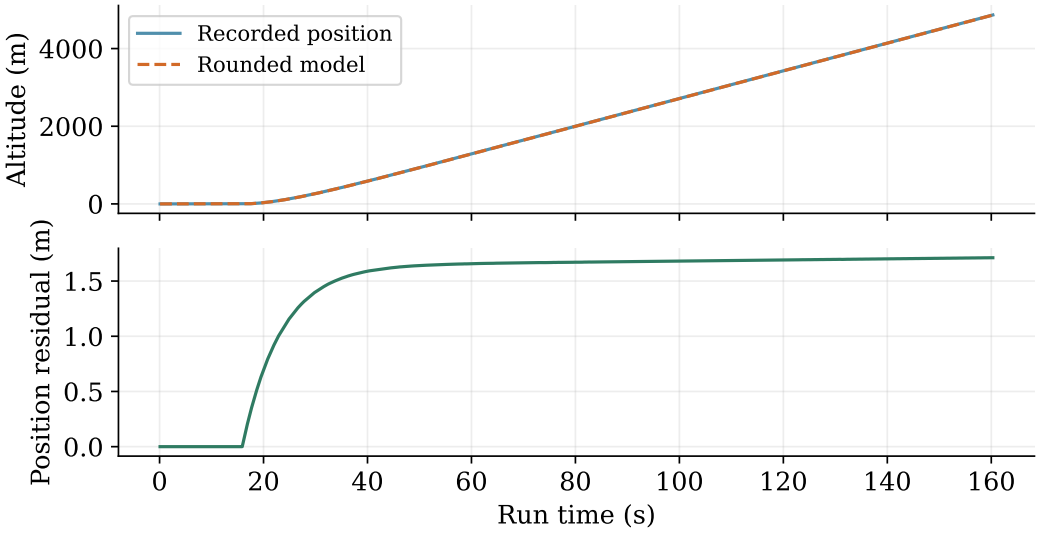


Figure F.1: Position-domain model comparison and residuals for the retained identification run.

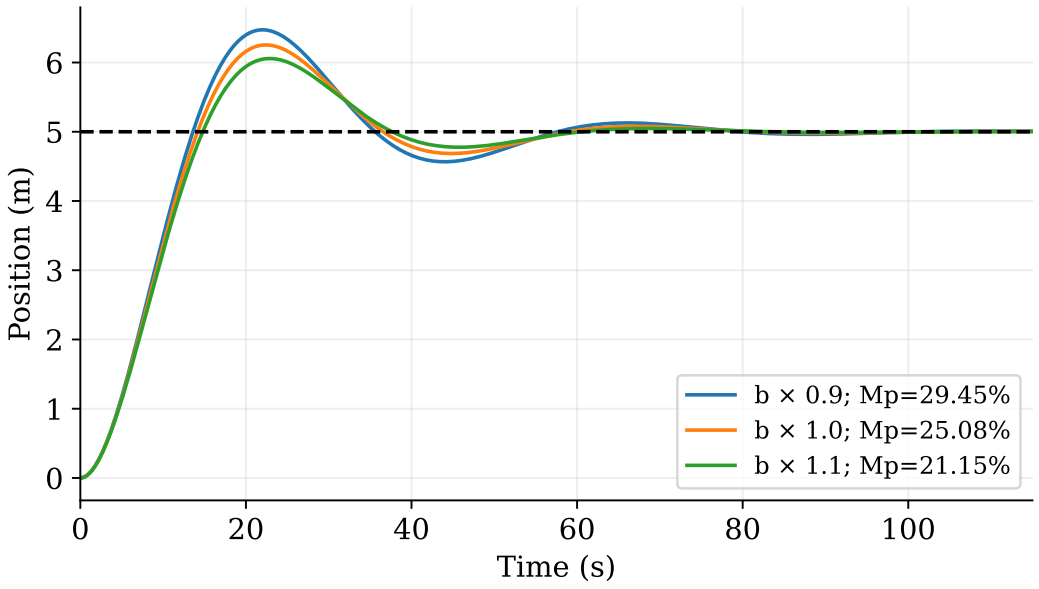


Figure F.2: Predicted altitude responses for $b = 0.9b_0, b_0, 1.1b_0$, with local thrust slope held fixed.

Figure F.3 demonstrates sensitivity to a defined input bias. At nominal gain its predicted final displacement is +0.150 m, within the 0.4 m allowance for a 5 m reference.

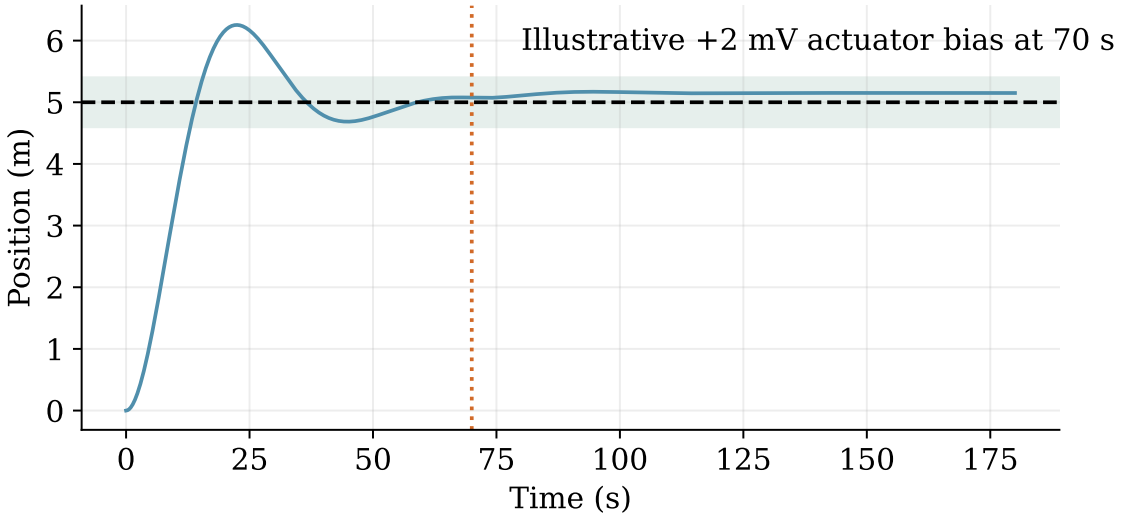


Figure F.3: Illustrative simulated response to +2 mV actuator bias added at 70 s. The shaded region is the 8% tracking band.

This simulation tests one stated disturbance amplitude. It does not imply rejection of arbitrary disturbance magnitudes or establish experimental disturbance performance.

G Identification Data and Verification Code

G.1 Data Provenance

The unmodified `data/session1/PTRJUN003_raw.csv` contains time, command voltage, sensor voltage, altitude and stored speed. The processed copy excludes the 31 pre-reset rows and adds recalculated velocity, midpoint time and model predictions.

Table E.1 gives selected calculations. The complete CSV and executable analysis reproduce all estimates; no previous-year data replace missing trials.

G.2 Organised Source Code

The numerical pipeline is separated by task. All stages read the shared parameter file; output CSVs and metrics are stored in results. Python generated the report figures. MATLAB equivalents require MATLAB with Control System Toolbox; the Simulink builder additionally requires Simulink.

Table G.1: Code modules and their calculation responsibilities.

Source location	Responsibility
Python modules	Identification, controller response/root locus, robustness, hardware digitisation and disturbance simulation.
MATLAB identification	Backward differences, velocity gain, 63.2% timing and sensor calibration.
MATLAB design	Root-locus geometry, angle/magnitude checks, gain limits and response plots.
MATLAB validation	Independent nominal and aerodynamic controller checks.
Simulink builder	Reconstructs and saves the nominal model.
Worked calculations	Separate calculation PDF, LaTeX equations and topic index.

Run the Python launcher from the project root:

```
python -m pip install -r requirements.txt
python analysis/analyse.py
```

In MATLAB, set the Current Folder to `matlab`, then enter:

```
run_all
% Or run only the detailed root locus method:
root_locus_design
```

The code guide records dependencies, file responsibilities and step conventions. Voltage identification uses a 2.5 to 5.0 V step ($\Delta u = 2.5$ V); the nominal controller test uses a 0 to 5 m altitude reference.

H Repository Navigation and Reproduction

Supporting files are available in the GitHub repository. Folder links open the corresponding folders on the main branch. Open the complete GitHub repository or use the folder links in Table H.1. Figure, table, equation and appendix references remain internal PDF hyperlinks. The repository guide and calculation index explain how to reproduce each result.

Table H.1: Clickable evidence map and corresponding Git folders.

Folder / evidence link	Contents
`data/session1`	Original identification CSV; source description in Appendix G.
`data/session2`	Original experimental display and Simulink screenshots.
`hardware`	Board photograph, LTspice/KiCad design sources and circuit notes.
`analysis`	Python identification, simulation, digitisation and tolerance calculations.
`matlab`	Controller verification and Simulink model builder.
`results`	Processed CSVs, nominal traces, corners and scalar metrics.
`report`	LaTeX source, figures and compiled report.
`scripts`	Build, link configuration and folder indexes.
`config`	One shared parameter file for MATLAB and Python.
`calculations`	Worked calculation PDF and equation source.

H.1 Build the Report

Extract the ZIP, keep its folder structure, install the Python requirements, and run:

```
python -m pip install -r requirements.txt
bash scripts/build.sh
```

The build regenerates numerical results and figures, then compiles `report/main.tex` with `latexmk -pdf`. A LaTeX distribution with TikZ and Circuitikz is required. The package also contains the compiled PDF for immediate viewing.

H.2 GitHub Folder Links

The report uses the repository supplied for this project:

NateDogg247 / EEE3094S—Lab-2-PTRJUN003-

The shared branch setting is `main`. Supporting-folder URLs use `/tree/main/`; individual source-file links use `/blob/main/`. `report/repository.tex` stores these settings centrally. Recompile the report after changing that file.

H.3 Page Count and Reproduction

The main report runs from Introduction on page 1 to Final Remarks on page 14. The title, table of contents, lists of figures and tables, references and appendices are excluded from this count. References use R-prefixed page numbers and appendices use A-prefixed page numbers.

The numerical pipeline regenerates identification, root-locus, response, tolerance and experimental metrics. The report and calculation supplement retain the corresponding worked equations. The available record contains one identification run and one controller display; it cannot establish repeated-trial averages or an assessor-awarded demonstration mark.